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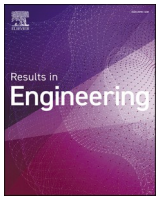
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Review article

# Advancing Sub-Seasonal to Seasonal Streamflow Forecasting in Canada: A Review of Conventional and Emerging Approaches for Operational Applications

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ABSTRACT

Sub-seasonal to seasonal (S2S) streamflow forecasts play a critical role in the planning and management of water resources for various purposes, such as optimization of hydropower production, ensuring sufficient water supplies for various usages, mitigating flood and drought risks, and management of nutrients from industrial and agricultural sources. Contrary to day-to-day operational activities, such forecasts can provide an extended operational window to various levels of the government for taking appropriate actions and issuing timely directives. Compared to the vast amount of hydrologic literature on short-term streamflow forecasting, S2S forecasting area is still not well-developed. This paper reviews state-of-the-art in S2S streamflow forecasting, considering conventional process-based and statistical modeling approaches, emerging machine learning (ML) techniques, and hybrid options. The generated knowledge and insights are intended to guide the development of operational tools for S2S forecasting for Alberta, Saskatchewan, and Manitoba provinces of Canada, and can also be used for developing similar tools for other regions of the world. Apart from discussing various modeling challenges, data availability constraints, and quantification of uncertainties, the paper also presents a systematic framework for developing ML-based S2S streamflow forecasting tools. Various limitations of the reviewed approaches and potential avenues of future research are also discussed to advance research and applications in S2S forecasting area. It is found that the potential of ML in addressing scaling issues in hydrology, through S2S forecasting, and investigating relevant hydrologic mechanisms at coarse spatial and temporal resolutions are not adequately explored. This is a significant path forward for ML in hydrology.

## 1. Introduction

Numerous decisions related to water resources planning and management activities in various sectors of society necessitate accurate forecasts of streamflow on sub-seasonal to seasonal (S2S) time scales, with a few months lead time, usually ranging from one to three months [1,2]. The S2S streamflow forecasting encompasses two distinct scales: sub-seasonal (typically 2–4 weeks) and seasonal (up to 3–6 months), both driven by extended-range meteorological forecasts. Although S2S terminology covers sub-seasonal to seasonal and monthly timescales, terminologies like seasonal and monthly streamflow forecasting are interchangeably used in this paper. In addition, this review focuses on S2S streamflow forecasting with 1–3 months lead times, aligning with seasonal partitions that are critical for planning and management in

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Canada and globally, such as spring runoff and water allocations for different purposes.

The S2S streamflow forecasts can help in making effective plans for meeting diverse water needs of various sectors, even during periods of water scarcity [3,4]. For agriculture, accurate seasonal forecasts enable farmers to plan irrigation schedules more efficiently, thus optimizing water use and improving crop yields [5]. In industrial contexts, reliable water supply forecasts are essential for processes that depend heavily on water availability. For domestic water supply, seasonal streamflow forecasts help water utilities plan for peak demand periods and manage water distribution systems more effectively [6]. According to Crippa et al. [7], reservoir operations and management can be significantly enhanced through improved seasonal streamflow forecasting [7]. Accurate predictions allow for better decision-making regarding water storage and release schedules, optimizing the balance between water supply for various uses, and flood control. This can help prevent water shortages during dry periods and to deal with excess water during wet seasons, contributing to more stable water management systems and enhancing the resilience of water infrastructure to various stressors. Unlike short-term predictions, S2S forecasts offer an extended operational time window for issuing sensible alerts. Nevertheless, achieving high forecast accuracy on longer timescales remains challenging due to limited inherent predictability in the atmosphere and hydrosphere, coupled with the constraints of model quality, complexity of the climate-hydrology interactions, and reliability of observational data [8].

Various studies have demonstrated that accurate and timely streamflow forecasts can result in significant benefits, although these benefits can differ greatly. For example, Rheinheimer et al. [9] reported a 1.2% increase in economic gains for the Sierra Nevada hydropower system. Turner et al. [10] examined the complex relationship between seasonal forecast skill and value in reservoir operations, revealing that while there is a strong correlation between forecast accuracy and operational value for reservoir management to maintain target water levels, this relationship is less straightforward for those aiming to meet specific water supply targets. In a broader analysis, Lee et al. [11] conducted a global analysis demonstrating that most dams (94%) could benefit from perfect forecasts, although the extent of these benefits varies based on reservoir characteristics and forecast skill. These insights underscore the critical role of integrating reliable seasonal forecasts into hydropower management strategies to optimize economic returns and ensure sustainable operations. Balancing energy production with ecological and social requirements is another crucial application. This balanced approach can lead to sustainable energy production, environmental conservation, community well-being, among many other benefits.

Generally, three primary approaches are employed for seasonal streamflow forecasting: process-based hydrological modeling, statistical modeling, and machine learning (ML). These approaches utilize information from historical and forecasted seasonal meteorological data to forecast future flows. The accuracy of seasonal forecast largely relies on the availability and continual improvement of various data types, the understanding of causal mechanisms and drivers, and modeling tools [12]. These approaches are discussed individually in greater details in Section 2 and Section 3 of the paper.

The earliest methods of seasonal streamflow forecasting were based on statistical analyses, driven mainly by regression techniques that leveraged prior hydrological conditions such as snowpack levels and soil moisture conditions to predict streamflow sequences for subsequent months [13,14]. These methods are still widely used even today, often involving complex regression models and teleconnection indicators [15]. It is important to highlight that statistical regression models continue to be the predominant data-driven approaches used in seasonal streamflow forecasting. They form an essential part of the core framework of most global operational prediction systems, generally favored for their efficiency and cost-effectiveness [16]. Process-based seasonal streamflow forecasting systems have also been used to offer both retrospective and operational forecasts. In 2019, the European Centre for Medium-Range Weather Forecasts (ECMWF) launched the fifth-generation seasonal forecasting system, SEAS5 [17], for seasonal hydrologic forecasting [18]. Hybrid systems that combine two or more different approaches (e.g., statistical and process-based modeling) are also in existence [19].

In the last two decades, substantial progress in information engineering and computer sciences has emphasized the promising role of ML in addressing complex problems in many scientific disciplines, including hydrology and water resources. ML algorithms are particularly noted for their capabilities in nonlinear modeling, making them highly effective for predicting streamflow at monthly and other temporal scales; a testament to their broad implementation is available in recent studies (e.g., [3,20]). Previous research has demonstrated the power and ability of ML in harnessing extensive datasets and decoding complex data patterns and relationships between predictors and desired streamflow outputs [21,22]. In spite of this notable advancement, the application of ML models for streamflow prediction is not without contention; the black-box nature of such models often results in outputs that lack clear linkages with physical processes involved in runoff generation within watersheds [23,24]. Pure ML models struggle with hydrometeorological conditions not seen during the training phase [25] and also face challenges in maintaining predictive accuracy under changing climatic conditions and changes driven by human impacts, which can alter hydrological patterns significantly, such as extreme events of precipitation and temperature [26]. In addition, uncertainty quantification of ML-based seasonal streamflow forecasting remains limited. Recently, there has been increasing interest in combining conventional process-based methodologies with ML models, capitalizing on the advantages of both techniques [24]. While there is a substantial body of research focused on process-aware ML [27], it is not yet comprehensive or wide-ranging enough to fully address the variety of processes found in hydrology, especially for S2S streamflow forecasting.

Despite notable progress in S2S streamflow forecasting, there remains a significant gap in the availability of comprehensive reviews that discuss the various challenges associated with conventional and emerging approaches. Previous reviews have focused on specific approaches to seasonal streamflow forecasting. Troin et al. [18] provided a critical review focusing on model-based ensemble forecasting approaches for streamflow forecasting. They highlighted the strengths and weaknesses of each approach, underlined the performance and uncertainty associated with various systems, and presented some key challenges and prospective future research directions. Easey et al. [28] contributed a short review on statistical methods and process-based modeling approaches in the context of seasonal streamflow forecasting. They suggested a method for combining both approaches.

In this paper, we cover a wide array of process-based and statistical approaches and also delve deeply into ML techniques and how these techniques can enhance or supplement conventional approaches. There is a noted absence of detailed reviews that focus exclusively on the application of ML techniques for seasonal streamflow forecasting. Therefore, this paper also attempts to fill this critical gap by discussing the application of various ML models, highlighting their strengths and weaknesses, and outlining the steps and components necessary for developing ML-based seasonal streamflow forecasting tools. One of the persistent challenges in seasonal streamflow forecasting is the quantification of uncertainty, particularly for ML-based systems. This aspect is also covered in this paper based on the published literature. In order to provide practical information to the hydrologic community, water managers, and end users, perspectives on developing ML-based seasonal streamflow forecasting tools that include important components of the forecasting systems, potential target areas of application, and important aspects in developing an operational system are also discussed. Although the paper covers both the international and Canadian literature and has global implications, many perspectives on advancing S2S streamflow forecasting and development of ML and hybrid models are discussed in

the Canadian context to guide the development of seasonal streamflow forecasting tools for Alberta, Saskatchewan, and Manitoba provinces of Canada, the focus area of this research. Fig. 1 provides a schematic depiction of various processes involved in developing an S2S streamflow forecasting system, including decision-making and target usages. The sequence and details of these components, described in different contexts in various sections of the paper, will likely vary depending upon the nature and complexity of a given project and target objectives of the forecasting system.

The remaining sections of this paper are organized as follows. Section 2 presents an overview of conventional approaches to seasonal streamflow forecasting, followed by an overview of ML approaches for the same in Section 3. A review of the quantification of uncertainty in seasonal streamflow forecasts is provided in Section 4. Perspectives on developing ML-based seasonal streamflow forecasting tools are discussed in Section 5, and finally, concluding remarks are provided in Section 6, including future research directions.

## 2. An overview of seasonal streamflow forecasting – conventional approaches

### *2.1. Process-based hydrological modeling and seasonal streamflow forecasting*

Through comprehension of hydrological processes and advancements in computer technology, numerical models for hydrologic forecasting were developed [15,29]. During the 1970s, one of the pioneering forecasting systems was established through the utilization of a hydrological model. This system was initialized with observed hydrological conditions and driven by historical time series data of recorded precipitation and temperature from past meteorological observations. This method was introduced by the National Weather Service (NWS) in the USA and was termed the Extended Streamflow Prediction (ESP) framework [29,30]. In the ESP framework [29], the hydrological model undergoes a spin up over a historical time period until it attains the initial conditions of the basin. Subsequently, the model is employed using resampled historical meteorological data to produce a set of potential streamflow forecasts for future time periods. Although the ESP demonstrates robustness, its reliance on historical meteorological forcing makes it impossible to capture hydrological responses to unprecedented meteorological events, never experienced before [18].

The scientific community has leveraged ESP forecasts to evaluate forecast skill sensitivity and uncertainties, and to benchmark seasonal forecast improvements [31] as well as for operational flood forecasting. Additionally, the ESP method primarily concentrates on the uncertainties in meteorological inputs and does not accommodate the non-stationarity of climate conditions [32]. To overcome these shortcomings, various modifications and improvements to ESP were proposed. One approach includes creating meteorological inputs for ESP by conditionally resampling past weather data based on the current states of climate indices (pre-ESP) [33]. Koster et al. and Mahanama et al. [34,35] employed a suite of land surface models, namely Variable Infiltration Capacity (VIC), and Catchment, Noah, and Sacramento. Their study encompassed a multidecadal dataset of meteorological forcings and time series of streamflow observations for 23 river basins, which varied in size from 2000 to 1.4 million km$^2$, covering a substantial portion of the USA. Their objective was to investigate the sources of streamflow forecast skill. They expanded this analysis through a 'synthetic truth' study that encompassed the entire area covered by the 48 conterminous United States (CONUS). This effort yielded a series of maps depicting the locations and timeframes where soil moisture and snow initialization contribute to streamflow forecast skill. Another common technique, known as trace weighting (post-ESP) [16], adjusts forecasted flow traces according to the similarity between climate characteristics of the forecast year and those of past meteorological years. More recent advancements incorporate predictions from climate models, enhancing the capacity to reflect future climate variability.

Risley developed a physical, process-based hydrologic model using the Precipitation-Runoff Modeling System (PRMS) in collaboration with the US Bureau of Reclamation for the Upper Klamath Basin [36]. This study demonstrated the capability of the model for forecasting seasonal streamflow volumes using ESP. The forecasted streamflow was derived using information about the exceedance probabilities of simulated streamflow, which was generated by ESP method and using multiple years of climate data. The results showed that the simulated inflow volume was very close to the 50 percent exceedance probability from the 10 forecasted scenarios. The study highlighted that employing a process-based hydrological model offers valuable insights into complex basins like the Upper Klamath, aiding in the development of seasonal streamflow forecasts. Research has explored the potential of using archived meteorological data and outputs from climate models to inform distributed hydrological models for projecting monthly/seasonal streamflow [37,38].

Over the years, various techniques have been developed to enhance the efficacy of forecasting systems. Crochemore et al. [39] used the GR6J hydrological model, the ECMWF System 4 precipitation forecasts, and eight conditioning methods to generate seasonal streamflow forecasts for 16 catchments spread over France. The authors employed four

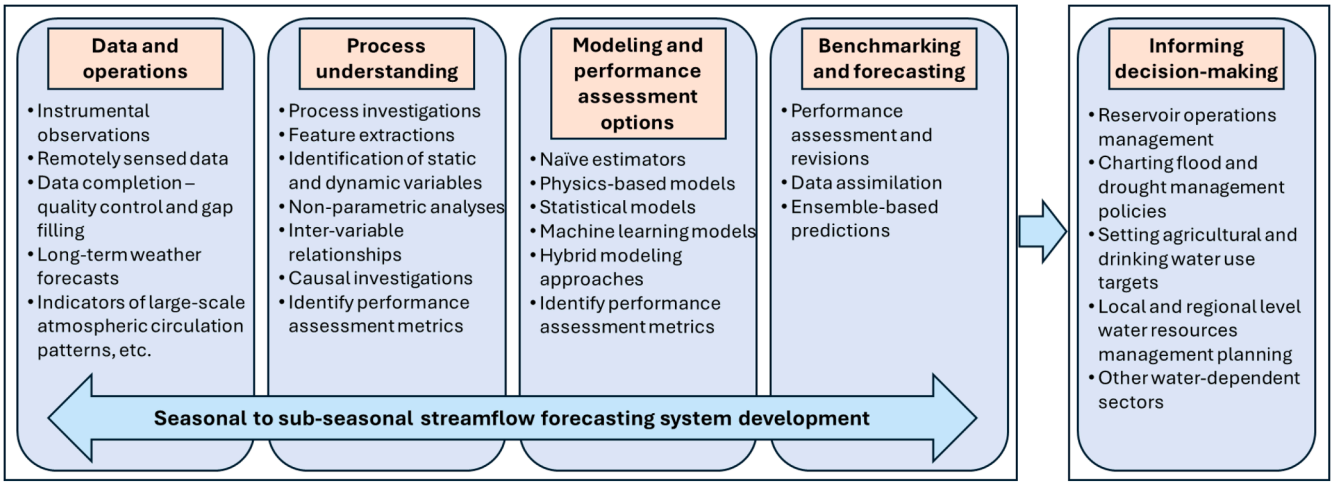



**Fig. 1.** A schematic depiction of various steps/processes involved in developing an S2S streamflow forecasting system, starting from data collection, feature extraction, process understanding, modeling options, and performance assessment and benchmarking to informing decision-making for various sectors of society.

conditioning statistics based on seasonal forecasts of cumulative precipitation and the standardized precipitation index to identify relevant signs in terms of historical streamflow and precipitation data. The results indicated that conditioned ensembles had increased reliability, however, exhibited fewer sharp forecasts compared to those generated solely from System 4 precipitation. Notably, in the context of low-flow forecasting, the conditioning approach yielded ensembles that were more adept at assessing weekly deficit volumes and durations across a wider range of lead times [39]. Arnal et al. [40] evaluated the effectiveness of using seasonal climate forecasts compared to historical meteorological observations for streamflow forecasting across Europe. The study utilized the European Flood Awareness System (EFAS), which integrates the LISFLOOD model forced with ECMWF System 4 climate forecasts to generate seasonal streamflow predictions. These forecasts are benchmarked against the ESP approach, which uses historical meteorological data as input to the LISFLOOD model. The findings indicate that while the ECMWF System 4 forecasts enhance predictability for the first month of lead time, their skill varies spatially and temporally, emphasizing the need for improved understanding of hydrological and meteorological interactions at seasonal scales.

Advancements in S2S streamflow forecasting are driven by improvements in algorithms and the availability of additional relevant data sources [41]. In 2019, ECMWF launched the fifth-generation seasonal forecasting system, SEAS5 [17]. This system has robust predictive capabilities, particularly regarding the El Niño Southern Oscillation (ENSO) [17]. SEAS5 can offer both retrospective and operational forecasts. The accuracy of streamflow forecasts was enhanced based on more precise initial conditions and meteorological inputs [18]. Streamflow modeling has benefitted substantially from the inclusion of refined land surface predictor data, with enhanced datasets encompassing critical geophysical variables, such as snow and soil moisture [42,43]. Girons Lopez et al. [44] conducted an evaluation of the operational ESP seasonal forecasts provided by the Swedish Meteorological and Hydrological Institute. They performed this assessment by benchmarking and attributing the forecast skill of ESPs over Sweden using the operational S-HYPE model. The outcomes of their research significantly contributed to identifying the specific geographical areas, seasons, and lead times into the future where ESP hydrological forecasts offered an enhanced level of utility. Presently, over Europe, multiple forecasting systems exist that also consider this improved predictive capability [40,45,46]. Most of these systems rely on meteorological seasonal forecasts generated by the ECMWF SEAS5 prediction system [17,47], incorporating various adaptations of a hydrological model. Hydrological models, such as LISFLOOD, LARSIM-ME, E-HYPE, SPHY, EVALHID, and wflow_hbv, were used in the seasonal forecasting systems. This deliberate model diversity was introduced to gain a deeper understanding of how the type of hydrological model influences the sensitivity of seasonal flow forecasts [45].

Process-based models are generally developed based on existing knowledge of hydrological systems and catchment-wide understanding of dominant hydrological processes, enabling them to effectively capture catchment dynamics and the underlying runoff generation mechanisms [48]. Process-based distributed hydrological models, such as wflow_hbv model, can account for spatial variability in precipitation, infiltration capacity, land cover, and topography within a watershed [1]. However, these models struggle to fully utilize diverse climate data, such as high-resolution reanalysis or ensemble outputs due to integration difficulties. Furthermore, they often have limitations in capturing complex nonlinear interactions (e.g., snowmelt timing, groundwater flow) because our understanding of these processes remains incomplete [49]. Additionally, employing distributed hydrological models for monthly streamflow predictions is computationally intensive and requires high-resolution gridded input data, particularly for larger watersheds, which can limit their practical applications [37,50].

### 2.2. Statistical and hybrid approaches for seasonal streamflow forecasting

Seasonal streamflow forecasting began with statistical approaches, particularly regression-based methods that predict streamflow for the coming months using antecedent hydrological conditions [14]. Pioneering work by Church used snowpack measurements—specifically snow water equivalent from surveys—to estimate spring runoff in snow-dominated regions, establishing a regression-based technique [13]. Their approach involved systematic snow surveys conducted at designated courses across the western USA. Linear regression was used to derive predictive equations, enabling seasonal forecasts of runoff. These early models correlated historical snow data with streamflow, a practice still valued in regions dependent on snowmelt due to its simplicity. Wilby et al. [51] explored the seasonal predictability of summer hydrometeorological variables—such as air temperatures, precipitation, soil moisture, and river flows—in the River Thames basin, UK [51]. Utilizing stepwise linear regression, the authors established significant relationships between the summer conditions and wintertime predictors, including sea surface temperatures (SSTs), sea-ice extent, and atmospheric pressure anomalies over the North Atlantic and Tropics [13,14]. The study highlighted the strong influence of SSTs on temperature and precipitation, sea-ice extent on precipitation and soil moisture, and pressure anomalies over the Barents/Greenland Sea on river flows, while also noting the role of antecedent winter conditions. Statistical forecasting methodologies remain prominently employed; for instance, Wang et al. [15] used an enhanced regression technique and the Bayesian Joint Probability approach to incorporate teleconnection indices—such as the El Niño-Southern Oscillation (ENSO)—for capturing large-scale climate patterns that influence seasonal rainfall and streamflow.

Based on the Bayesian decision theory for considering available hydrologic information, Krzysztofowicz and Reese developed a Bayesian statistical framework to characterize streamflow forecast uncertainty, enhancing operational efficiency for water management in the western USA [52]. Their approach, the Bayesian Processor of Forecasts (BPF), calibrates seasonal snowmelt runoff predictions by integrating historical data and forecast inputs, producing a posterior probability distribution of total runoff volume. They also introduced the Bayesian Correlation Score, linking forecast skill to economic value, and a statistical predictor of monthly cumulative runoffs, improving resource allocation during snowmelt periods. Slater et al. [53] proposed a statistical-dynamical strategy for generating probabilistic seasonal streamflow forecasts. This method integrates land cover data and precipitation forecasts from climate models. The approach encompasses the prediction of both low and high flows within the Raccoon River at Van Meter basin, an 8,900 $km^2$ basin situated in central-western Iowa, USA. By updating the relationship between the predictand and predictors annually, forecasts spanning from one to ten months ahead of each season were computed. The method combines a persistence forecast, based on previous year's annual row crop acreage, and monthly precipitation forecasts from eight climate models affiliated with the North American Multi-Model Ensemble. Borne et al. [54] employed an Ensemble Kalman Filter framework to forecast runoff at the sub-basin scale, utilizing openly accessible global datasets of reanalysis precipitation (ERA5-Land) and bias-corrected, spatially disaggregated seasonal forecasts (SEAS5-BCSD) for the São Francisco River Basin in Brazil. Runoff prediction was carried out through least squares estimation that capitalizes on the covariance patterns between runoff and precipitation. The effectiveness of this assimilation approach was evaluated using various ensemble skill metrics.

Hybrid systems that blend statistical and process-based methods have also emerged for seasonal streamflow forecasting. For instance, Rosenberg et al. [55] proposed a hybrid approach that leverages the strengths of both statistical and process-based methods to enhance seasonal streamflow forecasting. The authors developed hybrid models that combine simulated snow water equivalent (SWE) data from the VIC

model with statistical regression techniques, specifically principal components regression (PCR) and Z-score regression. These hybrid models utilized predictor variables such as SWE, accumulated precipitation, and antecedent runoff. The authors employed the VIC model to simulate streamflow, integrating it with statistical techniques like PCR and Z-score regression to generate forecasts. Findings indicate that VIC-based forecasts improve accuracy over traditional statistical methods, particularly for late-season forecasts when snow data is scarce, with skill scores showing enhanced reliability for snow-dominated regions like the Sierra Nevada and Colorado River basins.

According to Fleming et al. [19], regression-based prediction systems face technical challenges and require upgrading or replacement. While forecast horizons can be extended through the use of climate indices (e.g., ENSO) or seasonal-scale climate model forecasts as predictors in regression models, selecting appropriate variables and interpreting the resulting regression coefficients often require extensive statistical knowledge and applied experience [20]. This is because the relationships between the predictors and seasonal streamflow are often complex, nonlinear, and region-specific, necessitating careful model calibration and validation. Regression models incorporating these predictors have demonstrated dependable forecasts for a 1-month lead time [56].

### 2.3. Status of seasonal streamflow forecasting in Canada

Currently, Canada lacks a nationwide streamflow forecasting guidance system [57]. The task of operational streamflow forecasting, including flood forecasting, in Canada falls under the jurisdiction of individual provinces and territories and numerous municipalities and conservation authorities. These entities operate autonomously, exhibiting variations in terms of capacity, expertise, priorities, roles, and obligations. Consequently, they have established their own methods for data collection, assimilation, flood forecasting, and early warning systems. However, this segmented approach can result in a sluggish uptake of new technologies and methodologies [58]. The primary responsibility of river forecast centers throughout Canada is to formulate potential flood scenarios pertaining to river flows and lake levels [59]. Subsequently, they issue advanced alerts or warnings as necessary to ensure public safety. As a standard practice, this forecasting process entails employing recorded hydro-meteorological data, meteorological forecasts, and some watershed attributes. These elements are integrated with a precipitation-runoff or streamflow routing model to forecast flow rates or water levels for a specified future timeframe [60], ranging from hours to months. Some of the modeling approaches discussed below are generally used for short-term streamflow forecasting and simulation purposes, but they can also be adapted at monthly and seasonal scales with relevant pre- and post-processing of input and output data and appropriate adaptations in the selected model's structure. The information on short-term and seasonal forecasting is discussed together to demonstrate lack of studies on the latter, requiring advancing this specific area in Canada.

Hydrological modeling and streamflow forecasting systems that are used for Canadian watersheds are discussed here. These systems primarily rely on process-based hydrologic models and statistical regression approaches. Certain models are tailored to specific regions and are designed to address the unique characteristics of distinct watersheds. For instance, the WARNS (Water and Routing Numeric System) hydrologic model – an adaptation of the UBC Watershed Model by the British Columbia River Forecast Centre [61] – and CLEVER [62] are optimized for snowmelt-dominated, mountainous, and prairie landscapes. In contrast, other frameworks and models, such as Raven [63], Hydrotel [64], and WATFLOOD [65,66], are more commonly applied in eastern provinces, where watersheds feature highlands, farmlands, wooded expanses, and urbanized locales. The developed models typically offer lead times up to 15 days [60].

In many parts of Canada, seasonal streamflow forecasts primarily rely on regression techniques that utilize present and/or past conditions as predictive factors [67]. Alberta Environment and Protected Areas publishes, through its River Forecast Center (RFC), monthly water supply outlooks between February and August for use across Alberta, which include seasonal weather forecasts and runoff forecasts for southern and central mountainous rivers. The outlook forecasts are prepared using principal component analysis (PCA)-based statistical models. Saskatchewan's Water Security Agency issues maps of potential runoff outlook across Saskatchewan on March 1$^{st}$ and April 1$^{st}$. The underlying analysis factors in antecedent soil moisture, snow course/snow pillow data, precipitation, and temperature to estimate potential runoff [68]. Manitoba's Hydrologic Forecast Centre uses the ESP system, similar to the one used by the NWS of USA. In British Columbia, the Columbia River Treaty mandates that BC Hydro generates seasonal runoff forecasts for the inflow of the Columbia River into the Mica Reservoir and other treaty projects. These forecasts are based on a linear regression model [69]. Acknowledging the impact of climate anomalies, like the ENSO and the Pacific/North American (PNA) pattern on the yearly variation of snowpack within the Columbia River Basin [70], Hsieh et al. [71] utilized a linear regression model that incorporates local precipitation and indices of ENSO, PNA, and the Pacific Decadal Oscillation (PDO) to forecast streamflow of the Columbia River at Donald for April to August period. BC Hydro also incorporated ENSO data, alongside preceding watershed conditions, in their regression-based water supply volume projections for specific reservoirs.

In Ontario, the Ottawa River Regulating Committee provides expert inflow and water level forecasts with lead times typically ranging from days to weeks for the spring freshet [72]. Ontario Power Generation uses hydrological models, such as WATFLOOD, for short-term predictions in their operational watersheds. All models are forced by weather forecasts from the Ministry of Natural Resources and Forests and Environment and Climate Change Canada (ECCC), offering forecasts for lead times ranging from 24 hours to two weeks [73]. In Quebec, the Centre d'expertise hydrique du Québec under the Ministère de l'Environnement, de la Lutte contre les changements climatiques also utilizes hydrological models, such as HYDROTEL, and weather forecasts from ECCC to generate streamflow forecasts with lead times up to 7 days [74]. Apart from the above presented information, it was not possible to access additional information on S2S streamflow forecasting for eastern regions of Canada through the published literature and personal communications.

Across the country, streamflow dynamics are largely influenced by snowmelt, glacier melt, rainfall, and groundwater flow, resulting in varied nival, pluvial, glacial, and intermediate streamflow patterns [75]. Many operational streamflow forecasting systems rely on precipitation-runoff models that might not adequately account for the intricacies of hydrological processes specific to cold regions [57], requiring additional development and research.

The above literature review indicates that current Canadian streamflow forecasting systems, characterized by select provincial and territorial approaches, are providing critical insights and water management capabilities; however, significant disparities in technological adoption and methodological integration and innovation still exist. The absence of a unified national streamflow forecasting framework might further complicate efforts to standardize responses to drought and flood risks and water management challenges. Most of the existing approaches focused mainly on short-term forecasts, ranging from hours to days, and utilized process-based models. Compared to this, ML-based forecasting systems for short-term, monthly, and seasonal streamflow forecasting are limited for Canadian watersheds and transboundary regions. Emerging trends in the application of advanced ML techniques, detailed in Section 3, offer promising avenues for overcoming some of these challenges, as demonstrated by their use in other regions, such as the USA, Asia, and Europe for enhancing S2S forecast skill. Noticeable research gaps include the underutilization of ML techniques for

hydrologic modeling and development of operational forecasting tools. In addition, a shortage of modeling alternatives exists for forecasting seasonal spring runoff volumes, despite their significance for various sectors and a range of applications, especially for large water bodies, such as Saskatchewan River Delta, Lake Winnipeg, and Hudson Bay.

## 3. An overview of ML and other emerging approaches for seasonal streamflow forecasting

Despite substantial progress in process understanding, hydrologic modeling still faces challenges ranging from structural inadequacies and parameter uncertainty of models to insufficient ability to assimilate information from big data. According to Beven [76], to address the complexities arising from heterogeneities and nonlinearities in hydrological processes, macroscale theories are essential across different catchment scales [76]. This issue remains a significant challenge in hydrology [23], which has opened up opportunities for unconventional approaches to address hydrologic problems. It is also argued that simple data-driven models suffer from lack of physics representability, transparency, and reliability [77]. To overcome some of these issues, advanced data driven models based on modern ML architectures have emerged.

ML-based approaches are increasingly being used for empirical data analysis in hydrology [78,79]. While ML methodologies have been utilized in hydrology for many years, recent computational advancements have allowed the organization of shallow learning models into complex structures with feature layers capable of learning multiscale patterns. This advancement enables the use of diverse data, including multi-catchment data, in interpretable ways [80]. At its core, ML comprises a set of techniques and tools for building and training models that can identify and replicate spatial and temporal patterns encountered in target datasets. The central philosophy of deep learning is to minimize human experts' intervention in feature design and maximize information extraction from the data [81]. According to Nearing et al. [24], large-scale hydrological datasets may contain significantly more information than hydrologists have been able to translate into theory or process-based models. As a result, the hydrologic community is increasingly harnessing the potential of available data using ML across various domains of hydrology [82].

Streamflow modeling and prediction has seen considerable advancements through the ongoing development of various ML architectures. Numerous studies have been published on the application of ML models to forecast streamflow magnitudes at various temporal scales, including hourly [83,84], daily [85–89], and monthly [90–92]. Although ML techniques, due to the nonlinear modeling capabilities, have attracted applications for monthly streamflow predictions [79,93], the available research on monthly and seasonal streamflow forecasting is relatively limited when contrasted with the abundance of literature focusing on sub-daily and daily time scales [78,94]. The process of streamflow prediction based on ML techniques involves multiple stages, including data pre-processing, feature extraction, algorithm selection, hyperparameter optimization, and model training and testing. In the following two sections, various ML approaches and architectures are discussed, followed by hybrid approaches and emerging trends to provide a reasonable overview of this area, especially in the context of S2S streamflow forecasting.

### 3.1. Commonly used ML approaches and architectures

Among the range of ML techniques, multiple linear regression (MLR) is frequently employed for long-term projections [95], but its assumptions are too simple. Since the earlier 2000s, artificial neural networks (ANNs), Support Vector Regression (SVR) were explored by Guo et al. [96] and Luo et al. [97], while those based on the implementation of random forest (RF) are evident in the works of Ahmadi et al. [98] and Akbarian et al. [56]. The k nearest-neighbors (kNN) method is another technique featured in the work by Costa et al. [99] and Khan et al. [100]. Genetic programming (GP) strategy has been pursued by Hadi and Tombul [101], Mehr et al. [102], and Mehr and Safari [103]. Tree-based boosting ensemble techniques, including eXtreme Gradient Boosting (XGBoost), have been explored in the studies by Akbarian et al. [56] and Ni et al. [104] for seasonal streamflow predictions. It must be noted that the structure of XGBoost is not without contention and therefore it must be used cautiously. Similarly, ML model structures that are not consistent with physical understanding of runoff generation in watersheds should also be used with caution. Since around 2017, Deep Neural Networks encompassing Long Short-Term Memory (LSTM) and Convolutional Neural Network (CNN), have seen strong popularity as explored by Ha et al. [105], Ni et al. [92] and Shu et al. [22].

In the discussion presented here, we focus on the application of ML models for seasonal streamflow forecasting. Various ML techniques have been explored for S2S forecasting, including LSTM networks [106] and their variants, which are well-suited for capturing temporal dependencies in streamflow data [88], and RF, which enhances predictive accuracy through ensemble learning [107]. Other methods, such as kNN [108] and GP [109], have also been applied for seasonal streamflow forecasting [110], offering non-parametric and evolutionary optimization capabilities. Table 1 provides a concise summary of the key ML models used for seasonal streamflow forecasting.

Recent research has showcased the superiority of LSTM, as a deep learning technique, over other structures for monthly streamflow forecasting [3,92,105,127,129]. For instance, Ha et al. [105] undertook a comparative analysis of three models, stacked LSTMs, Convolutional Encoder-Decoder LSTM (ConvLSTM), and Convolutional Encoder-Decoder Gated Recurrent Unit (GRU), using the ENSO pattern for forecasting monthly streamflow in Yangtze River Basin, China. Their findings indicate that ConvLSTM model enhances the robustness and precision of monthly streamflow forecasting. Additionally, the inclusion of ENSO in the experimental dataset led to more precise forecasts of high flows. Best outcomes were achieved through the employment of ConvLSTM combined with the Encoder-Decoder GRU. This architecture amalgamates several models, capitalizing on the strengths inherent in each of them. LSTM and GRU are utilized as decoders. However, the

**Table 1**
A summary of popular ML techniques used in seasonal streamflow forecasting.

| ML technique | References | Inclusion of weather forecasts | Number of basins considered | Basin areas ($km^2$) |
|---|---|---|---|---|
| ANNs | [111] | | 1 | 19,270 |
| | [112] | | 1 | 290,000 |
| | [113] | | 1 | 1,820 |
| | [114] | | 4 | 95,000–458,800 |
| | [115] | | 4 | 95,000–458,800 |
| | [116] | | 1 | 498,968 |
| | [56] | X | 30 | 6,827–226,523 |
| Support Vector Machines | [117] | | 1 | 24,959 |
| | [118] | | 1 | 754 |
| | [119] | | 1 | 239 |
| | [96] | | 1 | 1,808,500 |
| | [120] | | 2 | 258,948–312,812 |
| | [97] | | 1 | 473,200 |
| RF and tree-based models | [121] | | 4 | 875–6,900 |
| | [122] | | 1 | 1,000,000 |
| | [123] | | 1 | 5,266 |
| | [56] | | 30 | 6,827–226,523 |
| kNN | [124] | | 4 | 95,000–458,800 |
| | [115] | | 4 | 95,000–458,800 |
| | [125] | X | 12 | 332–290,000 |
| | [99] | | 2 | 12,400–20,700 |
| GP | [126] | | 2 | 75–1,440 |
| | [102] | | 1 | 20,000 |
| LSTM and its variants | [3] | | 2 | 11,950–26,300 |
| | [92] | | 2 | 860,000–1,488,036 |
| | [127] | | 6 | 1,165–297,850 |
| | [105] | | 1 | 1,488,036 |
| | [128] | | 2394 | 3–3,634,880 |

GRU stands out, owing to reducing parameter complexity compared to the LSTM, and that leads to computational resource savings. For six stream gages in the Murray Darling Basin, Australia, Masrur Ahmed et al. [127] combined an LSTM network with the Boruta-random forest (BRF) algorithm to forecast monthly streamflow water levels (SWL). Four types of antecedent lagged data including SWL, rainfall, periodicity factor, and climate mode indices were used to forecast one-month-ahead SWL. Their findings indicate that incorporating significant lagged inputs enhances the accuracy of SWL forecasts compared to non-BRF deep learning models that lack prior feature selection. The hybrid LSTM approach demonstrated distinct advantages in forecasting SWL, with over 98% of prediction errors confined within a range of ±0.015 m and achieving relatively low relative errors, surpassing the performance of all benchmark models. Ni et al. [92] developed two hybrid models, wavelet-LSTM and convolutional LSTM for 1-, 3-, and 6-month ahead streamflow forecasting with inputs of 12-lagged records. The results of the two hybrid models were compared with those of MLP and LSTM models. Their findings emphasize that LSTM is suitable for monthly streamflow and the incorporation of wavelet transforms and convolutional layers enhanced the forecasting accuracy of LSTM, particularly for predictions over longer lead time.

DT-based models, such as XGBoost and RF have also been utilized for seasonal streamflow forecasting. The models offer some advantages as they do not require any assumptions about the distribution of predictors. Ni et al. [104] used XGBoost coupled with Gaussian mixture model for monthly streamflow forecasting in Yangtze River Basin, China. The applications of RF for monthly and seasonal flow predictions remain relatively limited. Talukdar et al. [123] used RF to forecast seasonal flow patterns within the Punarbhaba River Basin situated on the Barind Tract spanning India and Bangladesh. Their findings indicate that RF showed good performance for forecasting seasonal streamflow. Akbarian et al. [56] adopted RF to forecast runoff over a 1- to 3-month horizon in Iran, leveraging the ECMWF ensembles of precipitation, runoff, and temperature over 30 basins in Iran. These forecasts were juxtaposed against outcomes from other widely used ML algorithms such as MLR, ANN, SVR, and XGBoost. Their findings demonstrated that RF has good performance among the considered models for monthly streamflow forecasting in arid and semi-arid climates.

Older ML techniques have also been used for S2S forecasting, e.g., Wu et al. [124] employed kNN and compared it with ANNs and autoregressive moving average model at intervals of 1, 2, 6, and 12 months ahead, focusing on two distinct stream locations in China. Their findings show that the performance of kNN is improved by using data preprocessing techniques of singular spectrum analysis and moving average. Furthermore, the kNN technique was adopted to enhance ensemble seasonal streamflow forecasts within the Upper Colorado River Basin. This was achieved by incorporating North American multi-model ensemble forecasts of temperature and precipitation at one- and three-month horizons, as well as historical streamflow data from the preceding three months, as weighting factors [125]. Karimi et al. [130] introduced the wavelet-Gene Expression Programming model (wavelet-GEP) to forecast both daily and monthly streamflow in the Filyos River, Turkey. Their findings underscored the heightened efficacy of the wavelet-GEP option when contrasted with ANN and adaptive neuro-fuzzy inference system-based models. Mehr et al. [131] suggested a hybrid model that amalgamated the multi-period simulated annealing optimizer with multigene genetic programming. This integration aimed to bolster the predictive capabilities of existing ML methods for forecasting monthly rainfall in heavily localized semi-arid regions. They suggested that the effectiveness of the proposed algorithm could be examined in forecasting both monthly streamflow and rainfall, particularly with extended lead times.

### 3.2. Hybrid approaches and recent trends

Both ML and process-based physical modeling approaches fall short of effectively capturing complex physical processes, such as hydrodynamics and varied precipitation-runoff generation mechanisms [32]. Recently, there has been a growing interest in integrating both approaches to leverage their strengths. Although research on physics-aware ML is rapidly evolving, it remains neither exhaustive nor sufficiently comprehensive to cover the broad spectrum of dynamical and physical processes, particularly those involved in runoff generation and moisture depletion within a watershed.

Physics-aware ML approaches can be divided into four groups based on how physical principles and understandings are integrated with ML, i.e., physics-guided ML (e.g., [132,80]), physics-informed ML (e.g., [133]), physis-embedded ML (e.g., [134]), and combination of both process-based and ML models (e.g., [135,32]). The first category has been applied to incorporate physical constraints into ML methods, such as through regularization techniques to enhance generalization capabilities while reducing the amount of required training data [132,136]. The approaches falling in the second category, which is also know as physics-informed neural networks–PINNs, are specifically designed to directly learn the space–time solution of problem-relevant state variables [137]. Specialized network architectures have been developed for PINNs to ensure adherence to specific physical laws, enhance training efficiency, and improve model accuracy [133,138,139]. In the third category, physics is embedded in modeling frameworks or modules in an end-to-end fashion. For example, Feng et al. [134] developed delta models (i.e., differentiable, physics-embedded hydrological models) that combine process-based models with LSTM-based parameterization. In this framework, process-based HBV model was integrated with ML for parameter estimation, achieving a balance between interpretability and predictive performance. The use of LSTM-based pipelines ensures that the model adapts to changing inputs, while the differentiable nature allows seamless backpropagation during training. In the fourth category, process-based models are combined with ML models [32,140]. Several studies have showcased practical applications of such hybrid approaches in hydrology. For example, Chadalawada et al. [141] developed a GP-based toolkit that automatically induces rainfall–runoff model structures using building blocks from multicomponent flexible modeling frameworks (e.g., SUPERFLEX). Building on this, Herath et al. [142] extended the methodology to semi-distributed rainfall–runoff models, enabling spatial representation of hydrological processes using GP, which further enhances the interpretability and realism of induced models. To our knowledge, there has been limited focus on seasonal streamflow forecasting using physics-aware ML approaches. In a recent study, Xu et al. [32] proposed a hybrid approach consisting of a simplified VIC model combined with a CNN and GRU to predict reservoir inflows 1, 3, and 6 months ahead in China. According to these authors, the hybrid models hold significant potential for monthly streamflow forecasting, especially where training data is limited.

The above-mentioned “physics-embedded” genre of physics-informed ML is also called “differentiable modeling” [80,143]. It seamlessly connects ANNs with process-based models and train them together in an end-to-end fashion on big data, flexibly allowing part of the system to be assumed as “priors” while learning part of the relationships from data. For example, ANNs can provide physical parameters or partial process representation to the process-based models. Thus, the process-based models serve as the connective tissue and indirect supervision is enabled; the need for observations for the direct outputs of ANNs is avoided. Differentiable hydrologic models have been tested at the continental [144,145] and global [146] scales for rainfall-runoff, soil hydrology [138], routing [147], ecological [148], and other processes. These models have demonstrated better spatial generalization than LSTM [134], as well as resolution of extreme events [132]. They can be configured to produce high-resolution simulations finer than the scale of observations and are already operational at continental scales [149]. These models can be used to learn missing relationships like impacts of spatial scaling on hydrologic dynamics. When fed with appropriate weather forecasts, differential models can be quite

relevant for S2S forecasting.

Future research in ML-based seasonal streamflow forecasting should also focus on investigating the scale issues and the possible interpretability of the results as well as the model structure and configuration. S2S forecasting is usually needed for medium to large scale watersheds with a few months lead-time, ranging from one to three months. The time window and the spatial resolution challenge the underlying physics of most process-based hydrologic models and rigorous analysis of the system under consideration. Process-based models generally operate on a daily time step, or even on finer temporal scales. For S2S streamflow forecasting, this means producing forecasts with 30 to180 time-steps lead time. This large window leads to deterioration of the forecast accuracy due to error accumulation. ML and traditional statistical models can be custom-built with a coarser temporal scale, allowing for higher forecast accuracy and thus opens the door for investigating and understanding the relevant hydrometeorological variables at large spatial and coarse temporal scales. ML-based models need to be extensively explored as double-edge tools that can improve modeling accuracy and address scaling issues in hydrology to understand various mechanisms at coarser spatial and temporal scales. ML models should be supplemented with model agnostic methods (e.g., partial mutual information) and causality analysis [150], and model-dependent methods to narrow down the list of potential predictors that can explain the scale-dependent hydrological mechanisms. In addition, hybrid models are often constructed by combining process-based models with ML techniques to leverage the strengths of both approaches. However, such frameworks are rarely applied for seasonal streamflow forecasting due to the limitations of process-based models in capturing long-lead dynamics. Therefore, the challenge of balancing physical interpretability with predictive accuracy can be addressed through causality-informed ML or physics-guided ML approaches. The causal inference analysis [150] offers a promising approach to maintaining physical relevance in purely ML-based streamflow forecasting frameworks by identifying and constraining input variables based on their causal relationships with the target variable. This helps preserve the physical meaning of predictors and improves transparency, while still leveraging the predictive capabilities of ML for S2S streamflow forecasting.

## 4. Quantification of uncertainty in seasonal streamflow forecasts

Despite notable advancements in S2S streamflow forecasting over recent decades, the forecasting skill remains constrained due to various sources of uncertainty. These include limited seasonal meteorological forecasting skills [151], inaccuracies in initial hydrological conditions (IHC), and errors stemming from limitations in numerical modeling and data processing [152,153].

Quantification of uncertainty in seasonal streamflow forecasting has primarily leveraged hindcast-based frameworks to assess the impact of IHC and meteorological inputs on forecast accuracy. Wood and Lettenmaier [154] pioneered a hindcast-based framework known as 'reverse-ESP', contrasting it with the conventional ESP approach. This framework aimed to evaluate the relative significance of uncertainties related to initial conditions and boundary forcing. In ESP, a hydrologic model, assuming perfect IHC, is driven by a forecast ensemble generated from observed meteorological data sequences. Conversely, reverse-ESP combines an ensemble of resampled IHC with a perfect meteorological forecast. Their findings indicate that this framework can quantify IHC and meteorological forecast related uncertainties, particularly in regions, such as northern California and southern Colorado. Building upon this framework, Wood et al. [153] extended previous investigations to develop the Variational Ensemble Streamflow Prediction Assessment (VESPA), which further assesses the impact of varying levels of uncertainty in different aspects on the uncertainty of streamflow forecasts by using a metric named 'skill elasticity'. This measure serves as a measure of the capacity to enhance seasonal streamflow forecasting skill by improving either the seasonal climate forecasts (SCF) or the IHC skill. Their findings suggest that improvements in SCF skill can significantly enhance streamflow forecast skill. Furthermore, the results emphasize the critical importance of advancing hydrologic modeling, expanding watershed observations, and utilizing data assimilation techniques. Arnal et al. [152] employed the VESPA method to measure the expected enhancements in the accuracy of seasonal streamflow forecasts stemming from feasible improvements, such as IHC and SCF. Recognizing the computational intensity of the original VESPA method, their study also introduced two computationally efficient approximations to serve as cost-effective alternatives. These methods were evaluated using a comprehensive dataset of 30 years of ensemble hindcasts for 18 catchments across the CONUS. Among these, the 'end point blending' method emerged as a particularly viable option. Their findings underscore the significance of selecting an appropriate verification score for conducting a goal-oriented sensitivity analysis.

Mazrooei et al. [37] proposed an innovative data assimilation technique, namely variational data assimilation (VAR-DA) method, for assimilating observed streamflow to correct initial conditions and assessing the utility of the method in improving the monthly streamflow forecasts from the VIC model, which was forced with monthly meteorological forecasts. The core idea of the method is minimizing the errors in predicting the streamflow by correcting the spatially varying VIC model's initial conditions. Their hypothesis is that addressing the two sources of uncertainty – correcting initial conditions and utilizing month-ahead meteorological forecasts from climate models – will provide improved monthly streamflow forecasts, particularly for months with limited skill in climate forecasts. The results demonstrated the role of VAR-DA in quantifying and reducing uncertainty of monthly streamflow forecasts by narrowed VAR-aided ensemble forecasts.

Common techniques, such as bias correction and ensemble recalibration, are widely used to refine raw meteorological data by eliminating systematic errors before driving forecasting models. Post-processing of input forecasts is critical for improving streamflow predictions in S2S forecasting systems. For instance, Emerton et al. [155] detailed the development of GloFAS-Seasonal, a global operational forecasting system, that applies post-processing to ECMWF seasonal forecasts to produce reliable streamflow predictions, demonstrating practical advancements in S2S hydro-meteorological forecasting. Crochemore et al. [156] showed that post-processed continental-scale seasonal forecasts can achieve performance comparable to catchment-specific models for 3-month streamflow predictions in 10 French catchments, underscoring the value of tailored post-processing for hydrological applications. More recently, Zhang et al. [157] developed an ML-based post-processing approach to adapt S2S precipitation forecasts for improving the quality of streamflow forecasts for four watersheds in the USA. Beyond model-based uncertainty quantification and post-processing of input forecasts, post-processing of S2S streamflow forecasts can further enhance their reliability and operational utility. In case of ensemble forecasts, statistical post-processing is generally employed to enhance forecast reliability and accounting for uncertainties.

In addition, some recent studies have also researched uncertainty assessments for ML models used for seasonal streamflow forecasting. Dehghan et al. [158] used Monte Carlo simulation for uncertainty analysis to quantify the accuracy of monthly streamflow forecasts from feed forward ANNs. They evaluated the performance of forecasts under varying input scenarios. Their uncertainty analysis demonstrated that the reduction of input predictors through PCA might have limitations in maintaining forecast accuracy as most of the forecasted values fell out of the confidence interval. Liu et al. [159] employed four MLR models to quantify the value added by incorporating GRACE total water storage anomalies (TWSA) for seasonal streamflow forecasting in four Chinese basins. The study also quantified the uncertainty of the MLR models. Not only do their findings underscore the utility of incorporating TWSA into seasonal forecasts but they also quantify significant improvements

brought about by these integrations, particularly in terms of reducing the uncertainty in model predictions across various basins in China.

The studies discussed above focus primarily on uncertainty quantification methods applied to S2S streamflow forecasting using physically based and ML approaches. However, several well-established methods—particularly Bayesian approaches and Monte Carlo techniques—have been effectively implemented for short-term forecasting (e.g., hourly to daily time scales) and could be adapted to S2S streamflow forecasting. Bayesian approaches, such as the Bayesian joint probability model [160], Metropolis–Hastings Markov Chain Monte Carlo algorithm [161,162], and Bayesian Model Averaging approach [163], are widely used to generate probabilistic streamflow forecasts and quantify forecast confidence by integrating prior knowledge with observed data. In addition, as discussed in [164], the application of advanced uncertainty analysis techniques, such as Monte Carlo simulations, is recommended to enhance the quantification and management of predictive uncertainties, thereby providing decision-makers with a more comprehensive understanding of the potential range of outcomes. In the context of ML, multiple ML techniques and methods, such as quantile regression forests [165], Monte Carlo dropout [166], and bootstrap aggregation (bagging) [167] have been employed to estimate uncertainty in hydrological forecasts. These methods can be adapted for ML-based S2S streamflow forecasting systems by integrating multiple ensemble inputs, retraining models on bootstrapped datasets, or interpreting predictive distributions directly from probabilistic models. Therefore, operational S2S streamflow forecasting systems may benefit from uncertainty quantification strategies that combine ensemble spread, Bayesian post-processing, and ML-based probabilistic inference to robustly capture uncertainty arising from different sources.

A significant limitation of ML studies is that most of them typically do not consider uncertainty quantification and generate only deterministic predictions [166,168]. Our review of the existing literature reveals that while numerous studies leverage ML techniques to enhance forecasting accuracy, fewer investigations specifically address inherent uncertainties associated with these models, especially in the context of seasonal streamflow forecasting. This crucial area deserves considerable research in the future, investigating how ML models contribute to, mitigate, or amplify uncertainties in hydrological forecasts. The applications of ML in hydrology, in general, and streamflow forecasting, in particular, are in thousands. However, the findings are case-specific, with limited general applicability. The superiority of a particular ML technique in one study might be offset in another one, and the argument in support of some important input variables in a few studies are refuted in others. The conflicting results in the literature might be normal, and usually are proven by correlation-type evidence. However, this approach contributes to the doubts raised by physical hydrologists against ML and make many water managers and policy makers reluctant to adopt such techniques. The absence of causality analysis widens the gap between data-driven and ML scientists on one side and pro physically based models on the other side. Causality analysis is critical to demonstrate if ML produces accurate forecasts, for example, for the right reasons, which builds users' trust and enhances our understanding of hydrological processes and the knowledge embedded in data.

## 5. Perspectives on developing ML-based seasonal streamflow forecasting tools

In this section, perspectives on developing ML-based seasonal streamflow forecasting tools are discussed, considering important components of the forecasting framework, potential benefits of fulfilling the forecast needs of different stakeholders, important aspects of an operational model, and provision of a generalized guidance for developing operational tools for Alberta, Saskatchewan, and Manitoba provinces of Canada, the focus area of this research effort. The information provided can potentially be used for other regions of the world to advance and formalize S2S streamflow forecasting area.

### 5.1. Important components of a seasonal streamflow forecasting framework

In general, a seasonal streamflow forecasting framework encompasses several critical elements that work in tandem to ensure accurate and reliable forecasts. First, data collection and preprocessing are paramount for developing a ML-based streamflow prediction system. The aim of this step is to build a dataset that includes all potential model inputs and the target outputs and to ensure that all potential input variables have equal representation, despite their differences [169]. Dataset building involves collection of monthly hydrological data (e.g., streamflow records and soil moisture and snowpack measurements); monthly meteorological data (e.g., precipitation, temperature, daylight duration, and teleconnection indices); meteorological forecasts (e.g., precipitation and temperature forecasts); and static watershed features (e.g., DEM, drainage features, and land cover and soil types). Incorporating remote sensing data from satellite observations can further enrich the dataset building effort [170]. Remotely sensed data may include snow cover information, terrestrial water storage anomaly, and vegetation indices. For example, the Soil Moisture Active-Passive (SMAP) [171] and Soil Moisture Ocean Salinity (SMOS) [172] are now available globally. Data from the Gravity Recovery and Climate Experiment (GRACE) and follow-on satellite revealed that terrestrial water storage anomalies can account for as much as 11% of the variability in intra-annual streamflow [173]. However, the integration of these satellite products for operational streamflow forecasting remains limited [170]. Practical challenges include long data latency (e.g., 45 days for GRACE), short record lengths, and coarse spatial resolution, which limit their applicability. The forecasting models using only GRACE data underperform compared to those using in-situ inputs [170]. These limitations underscore the need for further research into data assimilation, downscaling, and the use of low-latency remote sensing products. It is generally believed that the performance of ML models improves with the consideration of more data and therefore additional data sources and products should be evaluated as they become available.

Second, selection of input variables (or feature extraction) involves identifying and selecting the most relevant variables that influence the target variable, i.e., monthly or seasonal streamflow in the present case. Recent studies have conducted input variable selection based on two main algorithms, i.e., model-based algorithms and model-free algorithms [169]. Model-based algorithms address the input selection problem during the calibration stage. This approach involves calibrating multiple ML models, including those requiring different subsets of inputs and selecting the option that achieves the highest prediction accuracy. Model-free algorithms depend on the intrinsic information content of the available data to assess the relevance of inputs, without relying on an underlying model. The common model-free measures include autocorrelation/partial autocorrelation, mutual information, partial mutual information, minimum-redundancy maximum-relevance [174], and gamma test [175]. Besides these prevalent approaches, we strongly suggest using causal inference methods, such as convergent cross mapping (CCM), to distinguish correlation from causation for the identification and selection of input variables [150]. The causal analysis involves identifying and understanding the cause-and-effect relationships between potential predictors (input variables) and the target variable. By focusing on variables that have a causal impact on seasonal streamflow, rather than merely correlated variables, the seasonal forecasting model can achieve higher levels of reliability, transparency, and credibility. This approach ensures that the model captures the true drivers of hydrological processes, leading to more meaningful forecasts.

Third, selection of candidate model structures and hyperparameter tuning is another critical stage. Initially the available data is divided into subsets for model calibration (training/validation) and evaluation (testing). The array of model structures could include MLPs, RNNs, LSTM networks, tree-based models (e.g., RF), kNNs, GP, physics-informed neural networks, and other variants. More recent

advancements include Encoder-Decoder and Transformer models, which are particularly adept at handling sequence-to-sequence problems and spatio-temporal relationships [85]. Each model transforms input data into desired outputs using different functional forms and parameters that require calibration. The diverse range of model architectures offer varied strengths, allowing for tailored approaches in developing seasonal streamflow forecasting models. However, it is important to properly characterize the problem (based on the selected inputs and internal data structures) as a regression- or a time series modeling-type problem and selecting ML techniques with structures that are theoretically fit for the problem, ensuring that good results are obtained for the right reasons.

Fourth, uncertainty quantification is also a crucial component. For ML models, it involves generating multiple realizations of the dataset using techniques, such as bagging, boosting, and bootstrapping to evaluate uncertainties in model parameters and forecasts. Additionally, employing various ML techniques and architectures—such as LSTM networks, RF, kNN, and GP—produce multiple outputs. These outputs can be used to assess uncertainties related to the model structure [176]. By comparing the performance and consistency of different models, it becomes possible to identify the strengths and weaknesses of each approach, thus providing a comprehensive view of the uncertainties inherent in the forecasting process. This multi-model approach ensures that the final forecasts are robust and account for different sources of uncertainty.

Finally, the operational implementation of ML-based seasonal streamflow forecasting involves translating the developed models into practical tools that stakeholders and end-users can use for decision-making. This begins with ensuring adequate computational infrastructure to handle the data processing, model training, and operational forecasting needs. A critical aspect of implementation is developing a user-friendly interface that allows stakeholders, such as water resource managers, agricultural planners, and emergency response teams, to easily access and interpret the forecasts. In addition, continuous monitoring and updating of models with new data ensures that the forecasts remain accurate and relevant over time. By focusing on these operational aspects, the forecasting framework can provide timely, accurate, and actionable insights to manage water resources effectively and mitigate the risks associated with extreme hydrological events.

### 5.2. Potential benefits of fulfilling the forecast needs of different stakeholders

Although studies on seasonal climate/weather and hydrologic forecasting are available in the literature, there is a need for developing research-based guidance from basic theory development to practical implementation in order to fully realize the potential benefits of seasonal streamflow forecasting [177]. In the provinces of Saskatchewan, Alberta, and Manitoba, ML-based seasonal streamflow forecasting—offering one-to-three-month outlooks—can deliver targeted benefits for water management, agriculture, hydropower, and community resilience. Despite uncertainties that may deter decision-makers, demonstrating province-specific economic and social advantages can drive adoption and such forecasts are invaluable for long-term planning.

In water management, particularly for dam operations, estimating the costs and losses associated with different decisions and considering their economic consequences is critical [178]. However, most studies on the economic value of forecasts involve theoretical examples without quantifying expected costs and losses, or they have used economic value as an additional measure to assess the quality of seasonal forecasts [179]. There are only a few studies that directly assess the economic value of seasonal streamflow forecasts, and these often do not provide monetary values. Compared to the conventional approaches, ML-based forecasting tools are much faster to develop and operate, a feature that can help generate multiple ensembles in a rapid manner, and this, in itself, has an economic value. The Alberta's RFC supports water managers and irrigators in river basins (e.g., Bow and Oldman River basins) with monthly outlooks. ML-based forecasts could enhance these by rapidly generating ensembles, improving reservoir operations and irrigation planning. For instance, quantifying avoided losses from drought against forecast costs could convince stakeholders of their economic value. Similarly, Manitoba's Hydrologic Forecast Centre serves Manitoba Hydro and flood-prone Red River communities. Faster ML tools could refine ESP-based forecasts, optimizing hydropower generation and providing early warnings for wet seasons, reducing damages. In Saskatchewan, the Water Security Agency's runoff maps guide farmers and rural municipalities. ML forecasts could quantify spring runoff volumes, enabling precise water allocation for over 40,000 farms and cutting irrigation inefficiencies, a critical need given the historical drought impact. The proposed forecasting framework should be designed to be flexible and adaptable, allowing it to accommodate the varying decision-making timelines, priorities, and information needs of different stakeholders—ranging from agricultural planners and water supply managers to hydropower operators and water management authorities.

### 5.3. Important aspects of developing an operational model

Previously mentioned applications of ML models for monthly streamflow prediction have primarily focused on a limited number of watersheds. To establish their credibility, these applications should involve numerous watersheds from various climate regimes. The model's performance varies with watershed area and the amount of data used in calibration or training of the model [46,180]. In recent years, scale issues in hydrological modeling have garnered significant attention [181]. A latest study by Xu et al. [128] claimed that ML-based monthly forecasting models are more suitable for watersheds with large drainage areas, greater than 3,000 $km^2$. However, to the best of our knowledge, the investigation of scale effects on ML-based seasonal streamflow forecasting models remained limited to date. From an applied viewpoint, ML-based seasonal forecasting models are expected to be applicable at various spatial scales, from small catchments to large river basins, and for various climate regimes. For seasonal streamflow forecasting, multi-basin models refer to hydrological modeling approaches that integrate data and analysis across multiple river basins instead of focusing on a single basin. As highlighted by Pechlivanidis et al. [46], multi-basin modeling not only improves the attribution of prediction uncertainty but also supports comparative hydrology by identifying systematic patterns across diverse basins. Furthermore, it facilitates cross-boundary and large-scale hydrological analyses, enabling researchers to generalize findings and enhance the robustness of seasonal streamflow forecasting methods. The multi-basin or regional modeling approach can provide a practical pathway for addressing scalability challenges by enabling ML-based S2S forecasting models to generalize across diverse watersheds while retaining context-specific hydrological characteristics. This approach aligns with previous studies, such as Wood and Lettenmaier [154], which demonstrated the role of ensemble methods in distinguishing the relative influence of initial conditions and boundary forcings on forecast accuracy. To enhance forecast capabilities for large systems, it is imperative to advance scientific and technological methodologies and foster collaboration among multiple research communities [47]. However, global-scale forecasting faces numerous hurdles, such as inadequate data availability that matches the required temporal and spatial scales for practical use, and challenges in disseminating forecast information to global stakeholders [182]. Predominantly, research on seasonal streamflow forecasting at the continental level has been conducted in Europe [1,40], on national scales within countries like the USA [34,35], and on large regions like in South America [47]. Notably, these studies used physical modeling approaches, such as ESP and ECMWF systems. Applications of ML-based seasonal streamflow forecasting are still limited.

Integrating long-term weather forecasts, such as those from ECMWF, can significantly enhance the accuracy of seasonal streamflow forecasting models. These forecasts can provide critical variables for driving ML-based seasonal hydrological prediction systems, including months-ahead precipitation, temperature, humidity, and wind speed. However, only a few studies have explored the potential application of weather forecasts in this context. Also, there is a need to analyze the possibility of using pre-trained ML models, which are trained and tested in data-rich watersheds, to forecast seasonal streamflow in data-poor watersheds. These pre-trained models could be effectively used for ungauged watersheds. Mangukiya and Sharma [183] showed LSTM's advantage for flood prediction in data-sparse regions on a daily scale. Perera et al. [184] introduced a modified Generative Adversarial Network to predict monthly streamflow in an ungauged, data-sparse basin, achieving high accuracy and effectively capturing extreme events. To improve model generalizability in data-sparse or ungauged basins, one possibility is to develop ML-based seasonal streamflow forecasting tools for different homogeneous groups of watersheds, which enables model training on larger datasets from hydrologically similar watersheds and allows shared learning across regions with limited records. This strategy can facilitate the transfer of predictive capabilities to ungauged or poorly monitored watersheds. Developing mechanisms to select the most appropriate model based on historical performance under similar meteorological conditions can greatly enhance forecasting reliability. This approach involves maintaining a repository of model performances across various meteorological conditions and selecting the model that has historically performed best under current and forecasted conditions.

S2S streamflow forecasting is essential for planning and managing available water resources effectively, especially for regulated basins [185]. Nevertheless, ML-based models encounter difficulties in preserving predictive accuracy amidst shifting climate conditions and human influences, which are likely to significantly modify hydrological patterns in the future. These changes include variations in precipitation averages and extremes, evapotranspiration rates, and river discharge levels [26]. Considering the limitations of ML models under changing climatic situations, one practical solution is to periodically re-train or re-optimize model parameters and/or structures using updated observations as they become available. This allows models to adapt to evolving hydrological patterns and extreme events. As mentioned above, another strategy is to adopt a regional modeling approach, in which basins are grouped based on physiographic and climatic similarities. This allows the models to be exposed to a broader variety of hydrometeorological scenarios, enhancing their generalization capabilities and robustness to extreme events. Additionally, to better account for climate-induced variability, it is also essential to incorporate climate-sensitive hydrometeorological inputs such as seasonal temperature anomalies, drought indices, snowpack estimates, and teleconnection indicators. These inputs will ensure that the models remain sensitive to the physical processes most affected by climate change, supporting the development of robust S2S streamflow forecasting systems. Furthermore, more attention should be paid to developing an operational model that has the capability to produce real-time forecasts, with updating capabilities based on the latest data and information. The various stages and components of a ML model development process for operational S2S streamflow forecasting are shown in Fig. 2. This figure presents a generalized view and specific implementations could vary from one application to another and across different regions and watersheds.

Despite the promising capabilities of ML-based S2S streamflow forecasting tools, several practical challenges must be considered to ensure successful operational implementation. First, data integration poses a challenge, especially in real-time systems where forecasts rely on diverse sources such as reanalysis products, forecasted climate variables, and streamflow observations from multiple agencies. Developing automated pipelines for real-time data acquisition, quality control, and preprocessing is crucial for sustaining model performance. Second, creating user-friendly interfaces for forecast visualization and decision support remains an underdeveloped area. Tools must be designed with stakeholder needs in mind, including clear communication of forecast uncertainty, accessibility for non-technical users, and adaptability across sectors such as hydropower, agriculture, and water management.

### 5.4. *Provision of a generalized guidance for developing seasonal streamflow forecasting tools for Alberta, Saskatchewan, and Manitoba*

Developing operational tools for seasonal streamflow forecasting for the vast Canadian region comprising provinces of Alberta, Saskatchewan, and Manitoba necessitates a thorough understanding of its unique landscape, climate, and hydrological conditions that considerably vary

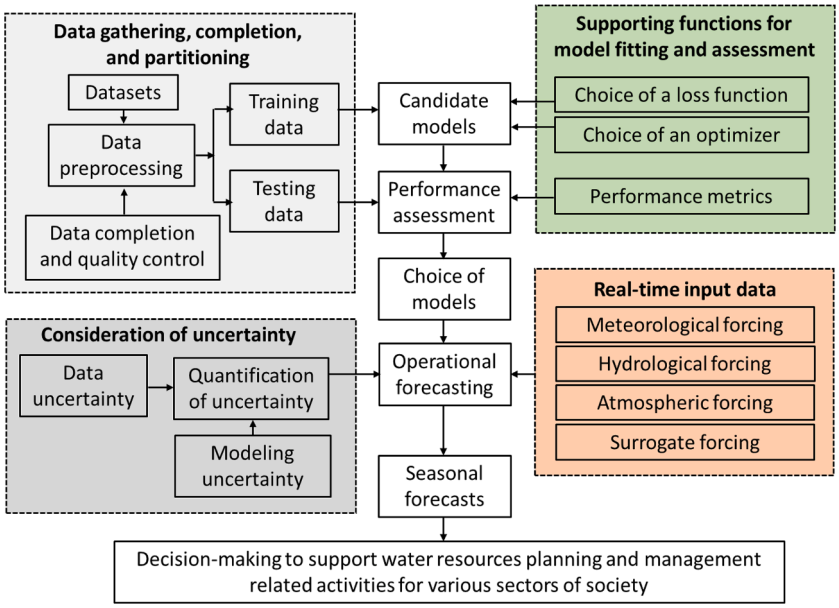



**Fig. 2.** A generic block diagram showing various stages and components of an ML model development process for operational S2S streamflow forecasting. Some components, actions, and processes are grouped together using different background colors to differentiate various stages involved. Arrows indicate sequence and relationships of various components.

across space. This region of Canada experiences significant seasonal variability influenced by snowmelt, glacier melt, and diverse precipitation patterns [186]. In mountainous areas, snow accumulation during winter and its subsequent melt in spring and summer are critical factors driving streamflow. Additionally, glacier melt also contributes to streamflow variability, especially during warmer months of the year in glacier-fed rivers [187]. Large portions of this region feature a unique landscape marked by potholes or sloughs. These areas remain disconnected during dry summers but contribute to flooding conditions through a fill and spill mechanism during wetter periods [188]. The extent of flooding can vary significantly, ranging from a few hundred meters to tens of kilometers across the plains because of the flat topography [189,190]. It is important to consider these dynamics and unique features into the development of ML-based seasonal forecasting models to produce reliable forecasts and ensuring model relevance.

While Canada's provincial and watershed-specific streamflow forecasting systems provide valuable insights, incorporating lessons from international practices can further enhance S2S streamflow forecasting capabilities. For instance, the EFAS integrates ECMWF SEAS5 forecasts with the LISFLOOD model to generate probabilistic seasonal streamflow forecasts across multiple countries, offering an approach for large-scale integration of climate and hydrological data [40]. In the U.S., regional ESP-based systems demonstrate the value of assimilating snow and soil moisture conditions into ensemble streamflow forecasting [35]. These examples highlight the benefits of centralized forecasting frameworks, ensemble methods, and climate model integration. In contrast, Canada's decentralized and provincially managed forecasting systems offer flexibility but may face limitations in harmonization and resource sharing. Therefore, developing operational tools in Canada could benefit from selectively adopting successful elements of international approaches, such as centralized data assimilation and multi-basin model integration.

Incorporating data assimilation techniques can further refine ML-based model outputs and their accuracy and reliability. These techniques integrate real-time observational data or surrogates into model simulations, continuously updating the forecasts as new information becomes available [191]. This approach can help correct initial conditions related errors and reduce seasonal forecast uncertainty, enhancing forecasting framework's performance over time [192,193]. For example, Metref et al. [194] assessed assimilation of snow observations for seasonal streamflow predictions. Boucher et al. [191] proposed an ensemble-based data assimilation approach by using ML for improving performance of the model.

In Canada, current streamflow forecasting systems are founded on process-based hydrological models and modeling frameworks, such as VIC, Raven, WATFLOOD, and MESH [57]. It is important to know that each of the building blocks of any distributed process-based hydrological model is itself a lumped model due to the difficulty of defining physical processes at extremely fine scales. Process-based models are adept at simulating the fundamental hydrological processes like snowmelt, glacier melt, and runoff generation, which are critical for accurately forecasting streamflow. ML-based models can handle such problems by identifying patterns and nonlinear relationships within datasets and between various variables. Integrating these approaches with process-based modeling can leverage the strengths of both methods, resulting in improved predictive performance [32].

Stakeholder engagement is pivotal for the development of effective operational tools for seasonal streamflow forecasting. Engaging with experts, local water authorities, municipalities, agricultural stakeholders, and indigenous communities ensures that the forecasting tools meet the expectations of those who rely on such forecasts. The co-development process helps in incorporating the local knowledge, experience, and specific requirements of various stakeholders. For instance, water resource managers need precise forecasts to optimize reservoir operations, while agricultural stakeholders require reliable predictions to plan irrigation schedules. Providing comprehensive training and ongoing support to end-users is vital for the effective utilization of forecasting tools. Training programs should cover how to interpret the forecasts, integrate them into decision-making processes, and use the forecasting tools effectively. This empowers stakeholders to make informed decisions based on accurate and timely availability of seasonal streamflow predictions, ultimately enhancing water management and disaster preparedness.

Encouraging ongoing research is critical to integrating the latest scientific advancements into ML-based forecasting systems. This includes incorporating new data sources, such as advanced satellite observations, high-resolution climate model outputs, and innovative hydrological data assimilation techniques. Research efforts should also focus on improving ML algorithms to enhance model accuracy and computational efficiency. In addition, implementing an adaptive management approach ensures that forecasting tools can respond to changing environmental conditions and evolving needs of various stakeholders. This involves continuously monitoring model performance, updating model parameters, and recalibrating models as new data becomes available.

## 6. Conclusion

This paper provides comprehensive perspectives on S2S streamflow forecasting, considering conventional process-based numerical and statistical modeling approaches, emerging ML models, and hybrid options. This was achieved through a systematic review of the literature and highlighting advancements, challenges, and gaps in each category of these approaches, emphasizing the need for continuous improvement and integration of new understandings and data sources based on critical reviews of published studies. Although globally applicable across many regions, the information provided here is gathered to guide the development of operational tools for S2S streamflow forecasting in Alberta, Saskatchewan, and Manitoba provinces of Canada in order to support various water management related activities, such as optimizing hydropower production, ensuring sufficient water supply for drinking and agricultural purposes, and managing the risks of recurrent floods and droughts.

Some progress has been made in applying ML models to seasonal streamflow forecasting. ML techniques, such as LSTM networks and their variants, have shown superior performance in capturing complex nonlinear relationships and improving forecast accuracy. However, the review highlights several challenges in model interpretability and handling diverse hydrometeorological conditions. It is found that there is a growing interest in physics-aware ML and combining conventional process-based numerical models with ML architectures to develop hybrid models by leveraging key strengths of both. Hybrid models can enhance predictive performance by incorporating domain-specific knowledge, established understanding of complex hydrological processes, and capturing hidden data patterns and inter-variable relationships.

In spite of the advantages of hybrid modeling, at this time the potential of even stand-alone ML techniques has not been fully utilized. The critical review of the current literature indicates that although many studies utilize ML techniques to improve forecasting accuracy, there is a notable dearth of studies focused on understanding the inherent uncertainties of ML-based seasonal streamflow forecasting. More importantly, the potential of ML techniques to address scaling issues in hydrology is significant, where hydrologic mechanisms and relevant predictors can be investigated at a large spatial scale and a coarser temporal scale to improve both S2S streamflow forecasting skills and hydrologic understanding. Such gaps highlight an important direction for future research for enhancing our understanding of scaling issues and how ML models reduce or exacerbate uncertainties in hydrological modeling, in general, and S2S streamflow forecasting, in particular. Advanced techniques such as ensemble forecasting and data assimilation are essential for assessing and considering uncertainties associated with modeling outputs, thereby improving the reliability and robustness

of forecasts.

From an operational viewpoint, perspectives on developing ML-based seasonal streamflow forecasting tools are also discussed in this paper. This includes important components of the forecasting framework and requirements of a real-time operational model. In addition, topics like data collection and pre-processing, input variable selection, selection of model architecture and model development, uncertainty quantification, and operational implementation are also discussed to provide a reasonable overview of the entire process of model building and operationalization.

## CRediT authorship contribution statement

**Duc Hai Nguyen:** Writing – original draft, Investigation, Data curation. **Amin Elshorbagy:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Muhammad Naveed Khaliq:** Writing – review & editing, Visualization, Funding acquisition, Conceptualization. **Chaopeng Shen:** Writing – review & editing. **Mohammad Khaled Akhtar:** Writing – review & editing. **Mohamed Moghairib:** Writing – review & editing. **Fisaha Unduche:** Writing – review & editing. **Saman Razavi:** Writing – review & editing. **Philippe Lamontagne:** Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

Data will be made available on request.

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